Joins in Relational Databases



Today's agenda



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Today's Questions

- How can tables relate in a relational database?
- What is partiality?
- What happens when I join along different types of relationships?
- Review

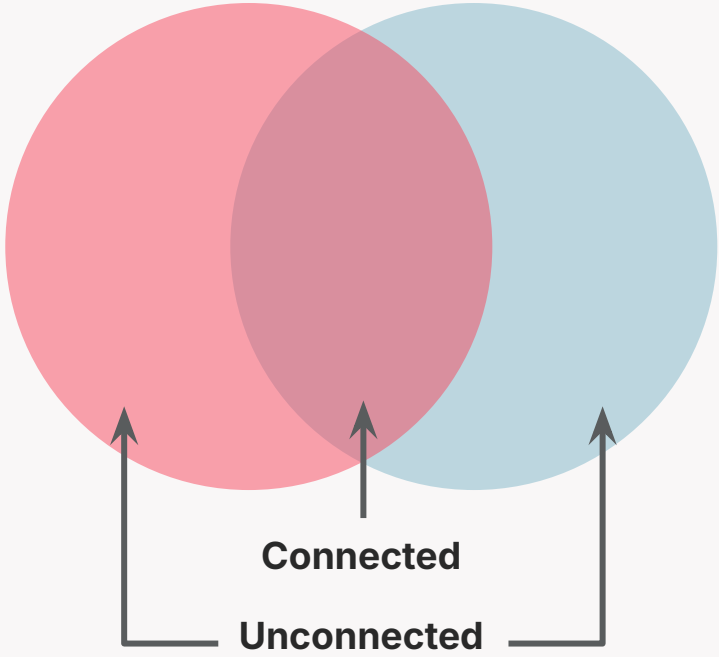
Joining Along Relationships | Partiality

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Joining Along Optional Relationships

In a “partial” or “optional” relationship, all joins produce different results. This is because there are **unconnected entries** from either table that could be excluded depending on the join.

Let’s take the example of a students table and a parking permits table. Students may or may not have a permit and permits may be assigned to students or non-students. How will the results differ with the joins?

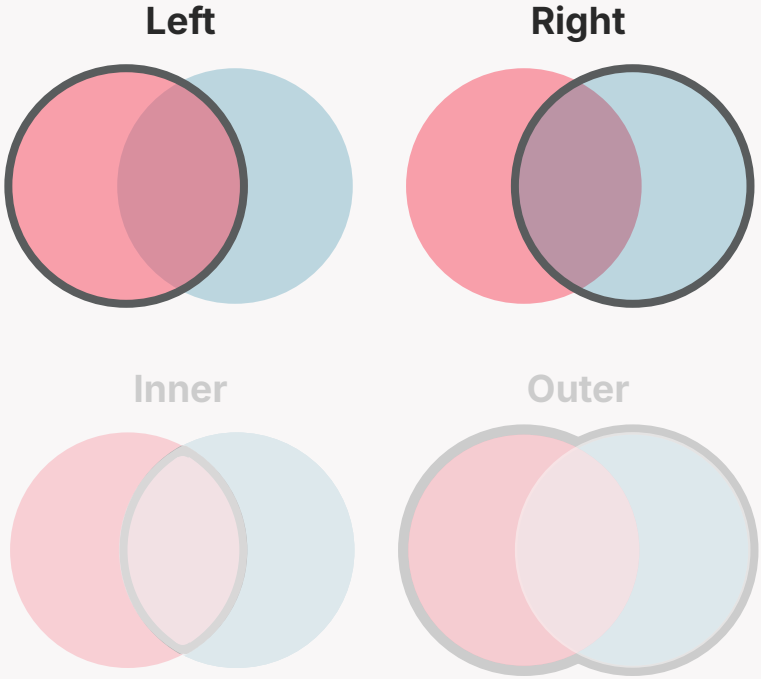


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Joining Along Optional Relationships

Left or **right** joins make one table "**dominant**", including all its entries and excluding all entries from the other table that are unconnected.



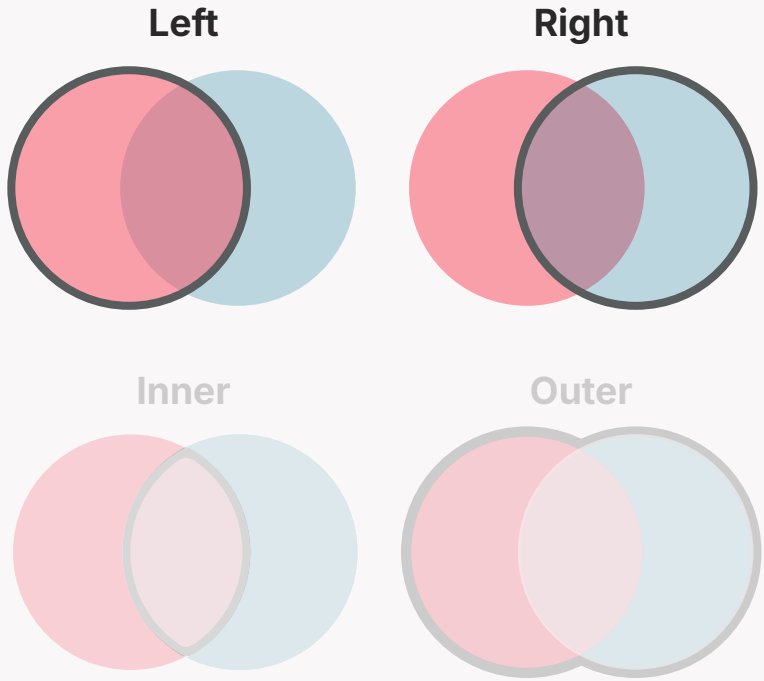
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Joining Along Optional Relationships

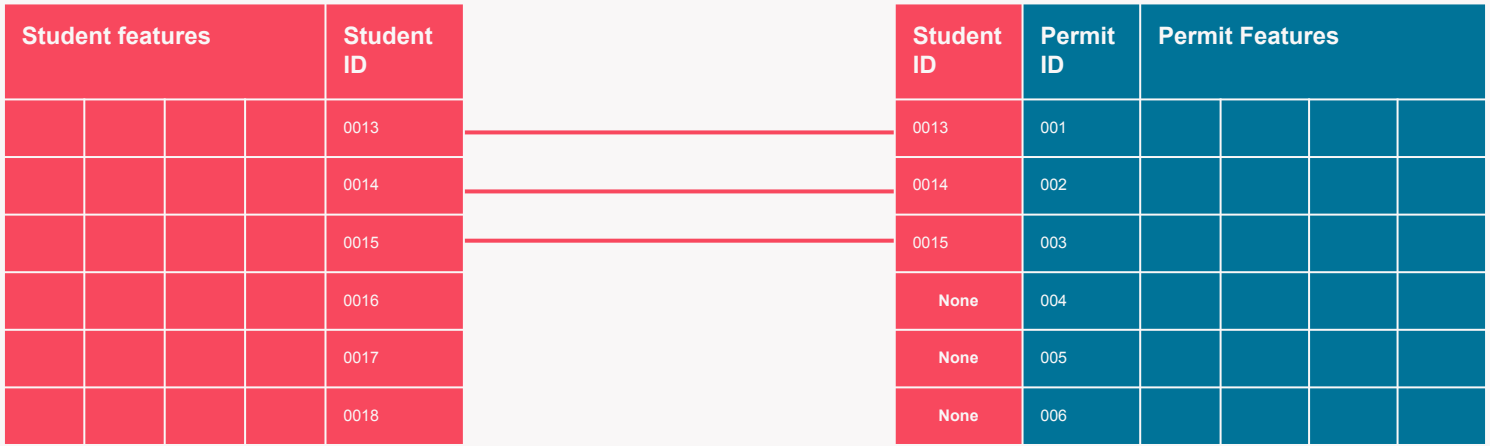
For instance, I may have the students table be dominant when joining with parking permits, resulting in the full table of students, now with parking permit information added to those students with permits.

You can think of this sort of join as *attaching* permit information to each student entry, or *expanding* each student entry to include more information.



Joining Along Relationships | Partiality

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Note: all partiality visualizations are of one-to-one relationships for the sake of simplicity

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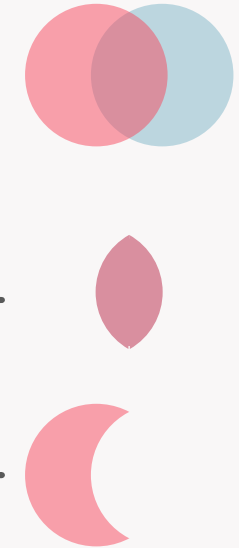
Student features				Student ID	Permit ID	Permit Features			
				0013	001				
				0014	002				
				0015	003				
				0016	None	None	None	None	None
				0017	None	None	None	None	None
				0018	None	None	None	None	None

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Joining Along Relationships | Partiality

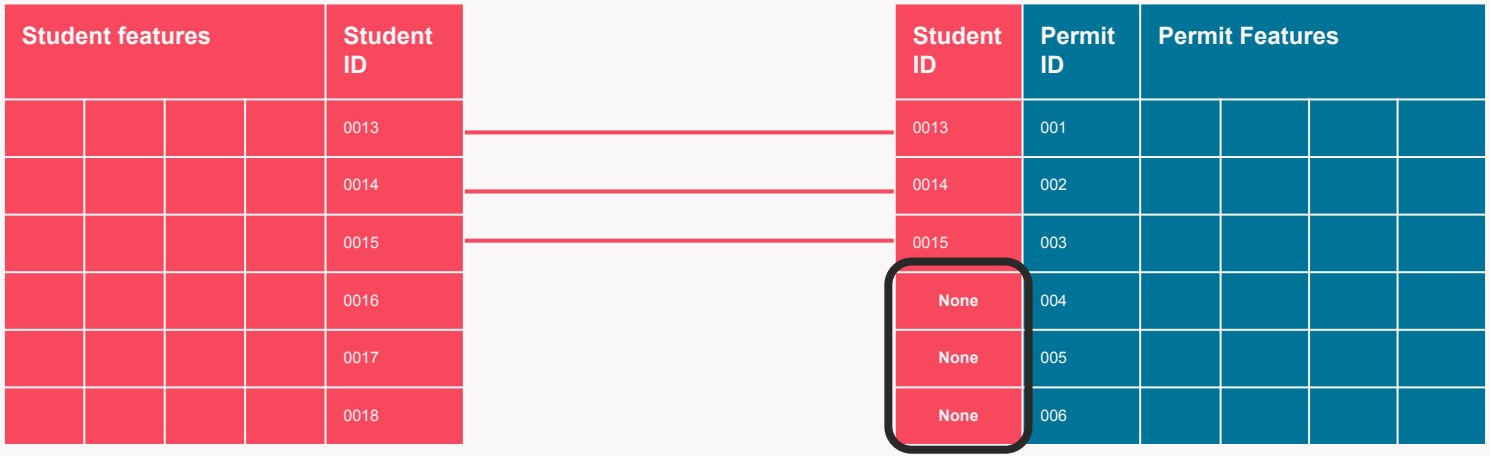
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Student features				Student ID	Permit ID	Permit Features			
				0013	001				
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				0015	003				
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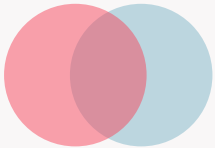
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Student features				Student ID	Permit ID	Permit Features			
				0013	001				
				0014	002				
				0015	003				
None	None	None	None	None	004				
None	None	None	None	None	005				
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Joining Along Relationships | Partiality

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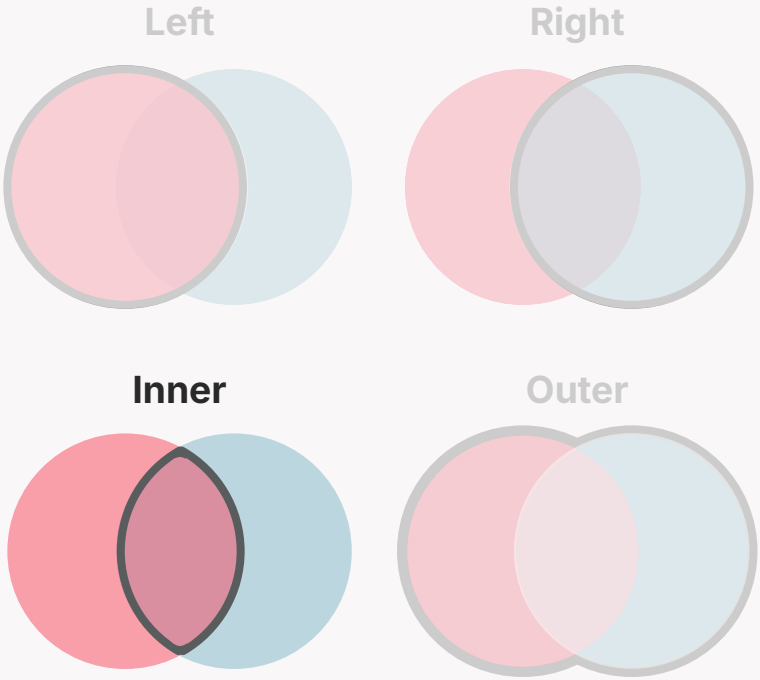
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Joining Along Relationships | Partiality

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Joining Along Optional Relationships

Inner joins will select only rows that connect the two tables.



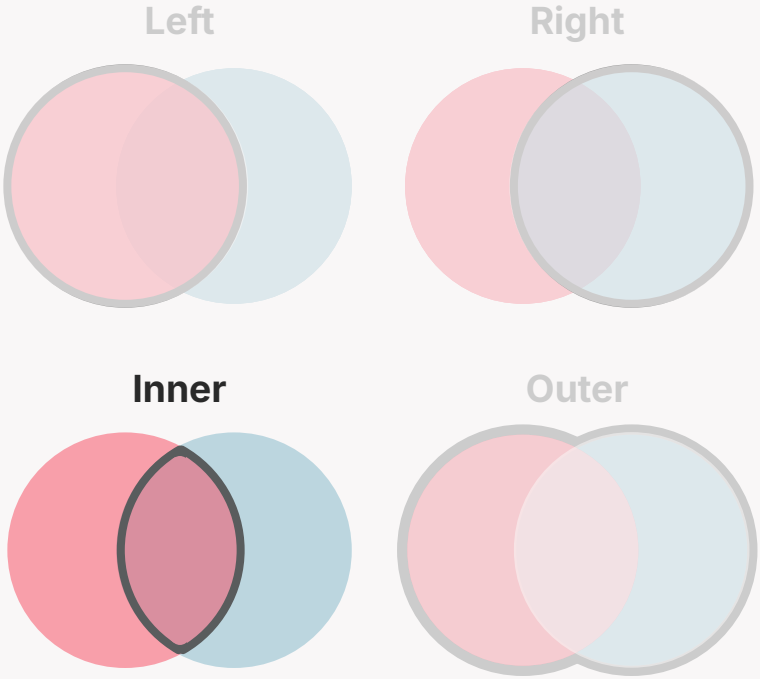
Joining Along Relationships | Partiality

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Joining Along Optional Relationships

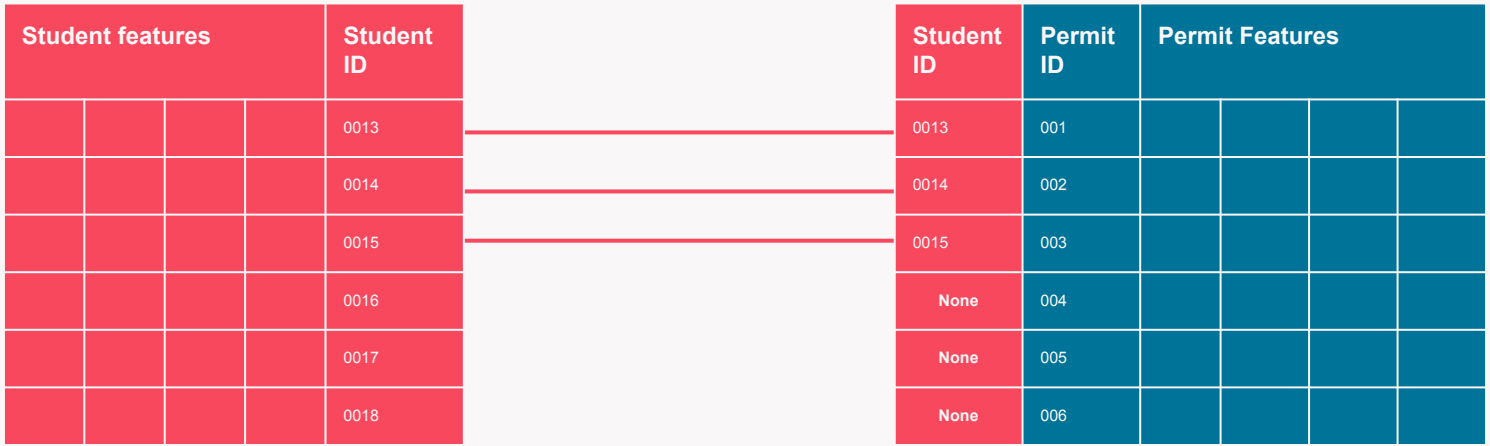
An inner join of the students and parking permits tables will only display permits that are owned by students and no information about students that do not have permits.

We can think of this join as *excluding* all entries that aren't involved in the connection between tables.



Joining Along Relationships | Partiality

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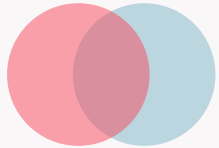
Joining Along Relationships | Partiality

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Student features				Student ID	Permit ID	Permit Features			
				0013	001				
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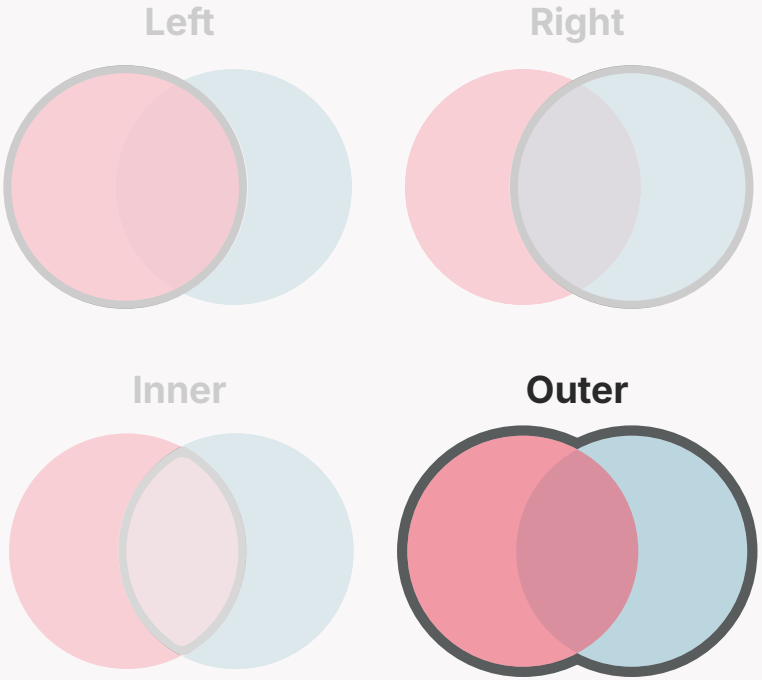
Note: all partiality visualizations are of one-to-one relationships for the sake of simplicity

Joining Along Relationships | Partiality

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Joining Along Optional Relationships

Outer joins will select all rows from both tables, excluding nothing.



Note: MySQL does not support the outer join SQL command. See appendix for how to do an outer join on MySQL

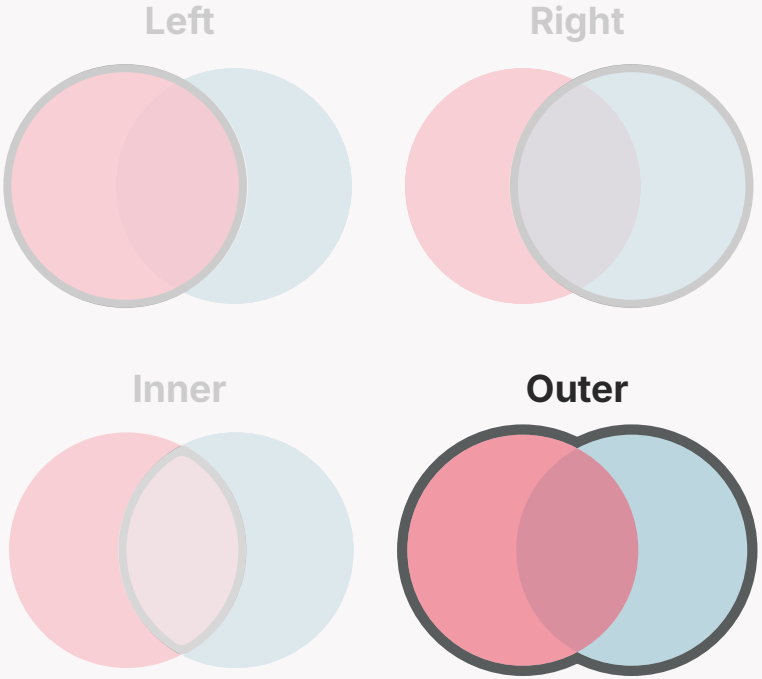
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Joining Along Optional Relationships

An outer join of the students and parking permits tables will display all the entries of both tables. The new table will have fields for both student information and parking permit information. Students without permits will have all permit related fields filled with “None”s and vise versa.

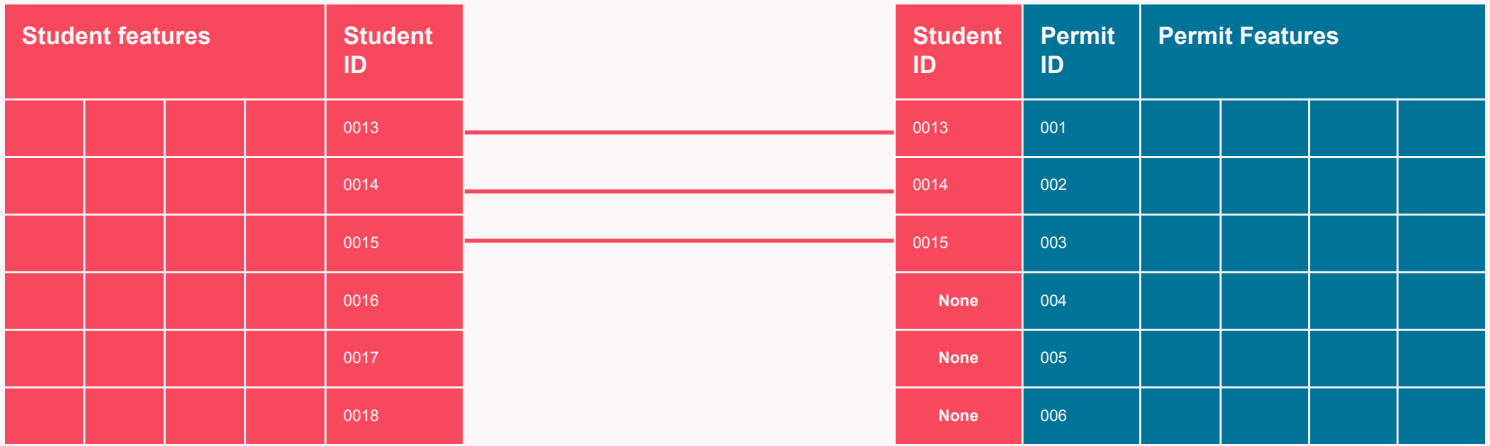
We can think of this join as *combining* both tables, connecting those entries that are connected and leaving the others with many empty cells.



Note: MySQL does not support the outer join SQL command. See appendix for how to do an outer join on MySQL

Joining Along Relationships | Partiality

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Joining Along Relationships | Partiality

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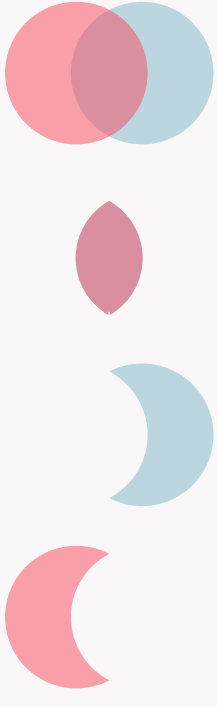
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Joining Along Relationships | Partiality

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Bonus: How to Do Outer Joins on MySQL

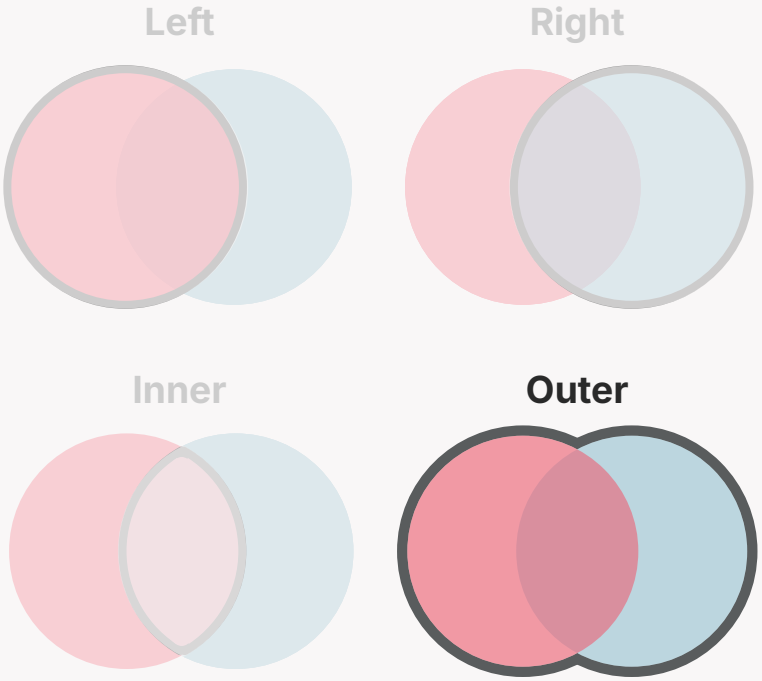
Hacks for Outer Joins on MySQL

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Outer Joins on MySQL

While the query language SQL does have syntax for outer joins, the RDBMS we use (MySQL) does not support outer joins.

Instead we will have to use a hack to accomplish the same functionality.



Hacks for Outer Joins on MySQL

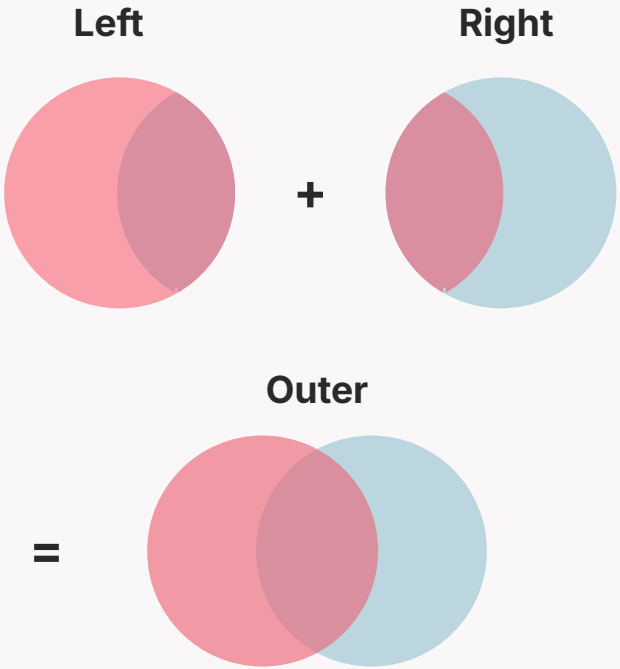
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Outer Joins on MySQL

The output we want should be a combination of all three parts of the venn diagram.

To get this output we can simply combine the output of a left join and a right join.

We can do this with a **Union** which will also **automatically remove duplicates** so that we don't duplicate the intersection that is shared by both the left and right joins' outputs.

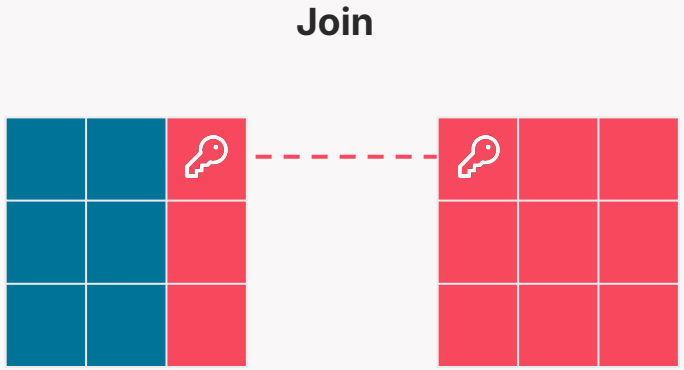


Hacks for Outer Joins on MySQL

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Unions

While joins unite tables *horizontally*, changing the number of columns (and often the number of rows as well), we can think of unions as concatenating tables *vertically* such that the number of columns stays the same but we simply add more rows to the table.



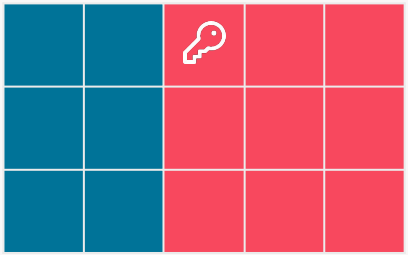
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Join



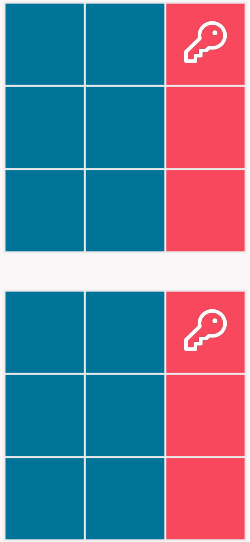
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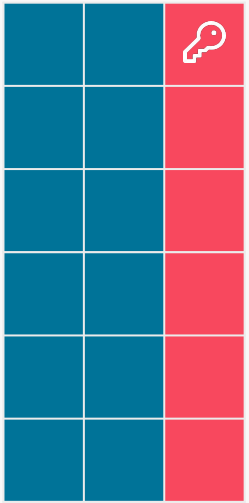
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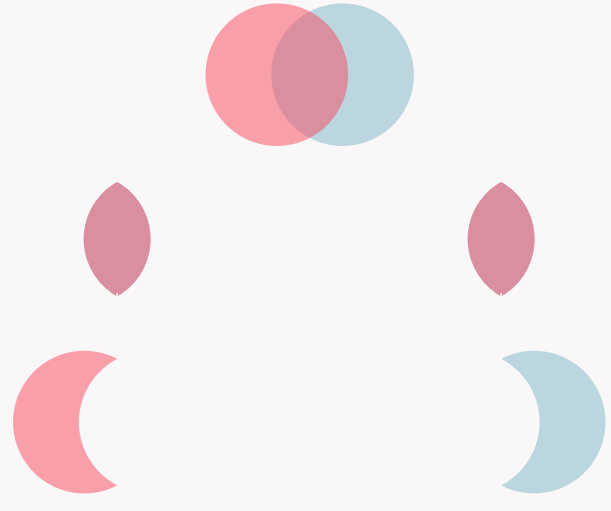


Joining Along Relationships | Partiality

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Left

nt	Permit ID	Permit Features			
	001				
	002				
	003				
	None	None	None	None	None
	None	None	None	None	None
	None	None	None	None	None



Right

Student features				Student ID	Permit ID	Per
				0013	001	
				0014	002	
				0015	003	
None	None	None	None	None	004	
None	None	None	None	None	005	
None	None	None	None	None	006	

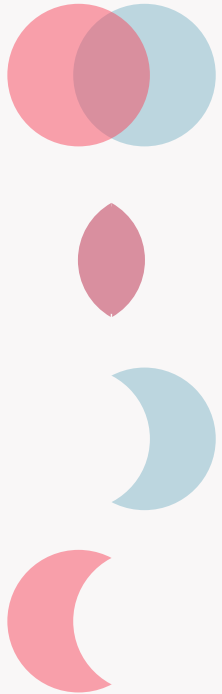
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Outer

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